

Arnab Mandal

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EDUCATION

Shiv Nadar University

B.Tech in Computer Science Engineering (8.59/10.0)

Noida, Delhi-NCR, India

Aug 2023 - Jul 2027

Heera Lal Public School, Class XII

Percentage : 91%

Madan Pur Dabas, Delhi-NCR, India

May 2023

Navy Children School, Class X

Percentage : 97%

Visakhapatnam, Andhra Pradesh, India

May 2021

INTERNSHIPS & EXPERIENCE

Intern, Enterprise AI Platform, Fidelity International, Gurgaon

Jun 2026 - Jul 2026

- Cut PII redaction latency by 92% via AWQ-quantized fine-tuning of Gemma 4 on SageMaker, reaching 96% F1.
- Shipped document sensitivity guardrail layer for enterprise AI gateway on 15k+ monthly uploads, audited via ELK.
- **Business Impact:** Cut manual audit effort by 40% and redaction costs by 85%.

AI Engineer Intern, DeepLure Research Pvt Ltd, Amravati

Dec 2025 - Jan 2026

- Optimized Llava 1.5 Pipeline serving 50+ clients by migrating inference to TensorRT + Triton.
- **Business Impact:** Achieved GPU utilization increase by 30% and serving 8x clients with no additional resources.

AI Intern, Critical AI Pvt Ltd, Bengaluru

Jul 2025 - Sep 2025

- Deployed multimodal on-prem RAG stack (FAISS, llama-cpp, FastAPI), cutting inference cost by 30%.

AI Intern, WESEE, Ministry of Defence, New Delhi

May 2025 - Jul 2025

- Cut SME report prep time by 90% via deploying 3+ MCP servers for agentic SQL analytics.

PROJECTS

Open-Source Contributor at vLLM (85k★): Domino Speculative Decoding 🧠🧡

- Implemented Inference and Distributed training support for Domino, reporting a 18% throughput gain on Qwen3.
- Improved acceptance length by 27% over DFlash across 7 benchmarks, including GSM8K and HumanEval.
- Technology Stack: Python, PyTorch, Transformers, FSDP2

Agentic Text2SQL Engine for Enterprise Healthcare Analytics Database 🧠🔗

- Engineered sandboxed multi-agent system with self-repair retry loop, tested over warehouse with 60+ relationships.
- Achieved 90% query execution accuracy, and validated by CEO+CTO for production adoption.
- Technology Stack: Python, LangGraph, FAISS, NetworkX, DuckDB, dbt, Airflow

AutoLinks: Distributed Semantic Internal Linking Engine for SEO 🧠🔗

- Architected async, autoscaling Goroutine workers with Redis-backed job queue + DLQ for bursty workloads.
- Utilized LLM-as-a-judge, achieving 85+% acceptance, with p95 Latency of 600ms, lowering Gini coefficient by 20%.
- Technology Stack: Go, chi, trafilatura, Qdrant, Redis

TECHNICAL SKILLS

Languages	Python, Rust, Go, TypeScript, C++, CUDA, SQL
AI & ML	LangChain, LangGraph, LangSmith, LiteLLM, Optuna, MLflow
MLOps & DevOps	vLLM, Triton, TensorRT, Ray Serve, Kubernetes, Docker, Jenkins, Prometheus
Cloud	GCP, AWS, Azure, Terraform

ACHIEVEMENTS

- **Smart SNU Hackathon'25 Winner:** Led a 6-member team to build and deploy the Krishi AI Sahayak app.
- **LeetCode Knight:** Achieved a 1,925 peak contest rating and solved 400+ DSA problems.
- **Semifinalist speaker at SMVIT Parliamentary Debate'25**